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Ayush Roy

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I am a second-year PhD student in Computer Science at Virginia Tech, advised by Dr. Doug Bowman and co-advised by Dr. Chris North. My research explores how people organize information, make sense of complex data, and collaborate in immersive environments, with an interest in how AI can further support these activities. I design and evaluate XR interaction techniques and interfaces through controlled experiments and mixed-methods user studies, with the long-term goal of making immersive computing a practical alternative to traditional desktop computing.

Education

Ph.D., Computer Science, advised by Dr. Doug Bowman — *January 2025 — Spring 2029(expected graduation)*
Virginia Tech

M.Eng., Computer Science — *January 2023 — December 2024*
Virginia Tech *(Transferred to Ph.D. program)*

B.Tech., Information Technology, — *August 2016 — December 2020*
Guru Gobind Singh Indraprastha University (GGSIPU), India

Publications

Understanding Asynchronous Collaborative Sensemaking Through a Multi-Session Exploratory Study in Immersive Workspaces — Present

A. Roy, C. Connor, S. In, K. Whitley, C. North, D. Bowman *(In Preparation for CSCW 27)*

- Developed an XR system for asynchronous collaborative data exploration with data logging pipelines for user behavior analysis.
- Conducted a multi-session mixed-methods exploratory study(N=30), combining quantitative behavioral analysis with semi-structured interviews and qualitative analysis to investigate **sensemaking strategies, challenges with handoffs, and collaboration patterns** as analysts built on each other's work.

Understanding AI-Assisted Organization in Immersive Computational Notebook — Present

S. In, J. Balayev, A. Roy, K. Whitley, C. North, M. Dasari *(Submitted to ACM CHI 2027)*

- Designed and conducted a **user study**(N=24) evaluating **3 levels of AI involvement** (Manual, Suggestion, Predefined) in organizing computational notebook cells, including assisting with semi-structured interviews to explore participants' experiences and strategies.

Characterizing Cognitive Load in Immersive Spaces to Think — Present

C. Connor, A. Roy, M. Baeck, S. In, K. Whitley, C. North, D. Bowman *(Submitted to ACM CHI 2027)*

- Contributed to study design and data analysis strategies for the study(N=24), investigating cognitive load through eye tracking, physiological signals, interaction behaviors, and subjective workload measures.

Evaluating Link Visualization Techniques for Sensemaking in Immersive Workspaces — March 2026

A. Roy, I. Tahmid, F. Rodrigues, S. In, K. Whitley, C. North, D. Bowman *(Submitted to IEEE VR 2027)*

- Designed and conducted a controlled 2×3 user study (N=35) evaluating **link routing and bundling** on graph tidiness and comprehension in immersive workspaces.
- Found that **many-to-many edge bundling and non-linear link routing** provide the most consistent benefits across tidiness, workload, usability, and completion time without reducing comprehension accuracy.

Exploring Organizational Strategies in Immersive Computational Notebooks — Oct. 2025

S. In, A. Roy, E. Krokos, K. Whitley, C. North, Y. Yang
IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)

- Co-designed the experimental study and conducted an exploratory user study(N=20) on spatial organization in immersive computational notebooks, identifying a preference for **half-cylindrical structures** and significantly greater engagement in non-linear analysis.
- Performed initial data analysis, identifying that participants increasingly adopted **multiple concurrent non-linear analytical approaches** as notebook size increased, informing design guidelines for immersive data science workspaces.

Selected Research Projects

Visualizing Collaborative Spatial Negotiation in XR — Present

- Leading the project in preparation for the **IEEE VR 2027 3DUI Contest**.
- Designing **dual-proposal visualization** and dynamic rubber-band-like feedback to communicate spatial disagreements, along with shared perspective to support negotiation and convergence toward a shared placement.

Multi-Agent Debiasing Framework with Online Search-Based Validation — Fall 2025

GitHub

- Designed **MABRF**, a multi-agent framework with a **two-agent core** (Judge + Rephrase) for automated detection and

iterative debiasing of LLM chain-of-thought reasoning.
- Integrated a Search Agent via DuckDuckGo API to retrieve real-world evidence before each rephrase pass; built a data pipeline across 110 BBQ-Hard instances and SQuAD v2, automating preprocessing, scoring (Exact Match, F1, Bias Score), and result aggregation.
- Evaluated four LLMs: baseline reduced mean bias score **from 0.062 to 0.008 (+3.7% accuracy)**; GPT-4 + search agent raised EM accuracy **from 27% to 45%.**

Understanding How Subtle Gaze Direction Cues Shapes Attention and Interpretation in VR Spring 2026
- Designed a within-subject VR eye-tracking study(N=18) with **6 cue conditions** (3 cue sizes × 2 luminance levels) to evaluate how subtle visual cues redirect attention without explicit user awareness.
- Analyzed **4 gaze and behavioral measures**: gaze hit rate, time-to-hit, button-confirmation rate, and pre/post gaze-velocity change; identified a **6-degree, 0.625-luminance** cue as the preferred balance between effectiveness and subtlety.
- Designed a follow-up **3-condition 360° narrative study** examining whether covert gaze guidance can influence character perception, trust, narrative understanding, and attentional integrity.

Professional Experience

Graduate Research Assistant *August 2024 — Present*
- **Immersive Space to Think:** Conducting research supported by the Department of War (DoW), developing immersive virtual reality (VR) tools to enhance intelligence analysis and analytical sensemaking.
- **DARPA ICS:** Designed and conducted a **mixed-factorial XR user study (N=20, 800 trials)**, supported by the DARPA ICS program, evaluating visual-search performance across **2 selection techniques** (Pinch, Gaze Dwell), **2 UI frames of reference** (Head-fixed, Hand-fixed), and **18 visual distraction configurations**.

Software Engineer, Infosys, *Pune, India* *March 2021 — January 2023*
- Automated financial computations via a data pipeline for spreadsheet uploads and real-time processing, reducing computation time from 60s to 200ms for **400,000+ records**; achieved **20% improvement** in efficiency via REST API integration.
  Tech: ASP.NET MVC, Angular.js, T-SQL, Azure DevOps.
- Recognized with the **Pfizer Champ Award** and **Rise Insta Award** in September 2021.

Software Engineer Intern, Credo India Softech Pvt. Ltd., *Delhi, India* *May 2019 — July 2019*
- Built a web application with REST API integration for customer analytics, achieving a **25% reduction** in support issues. **Tech: React.js, Node.js, Express.js, D3.js, Scikit-learn.**

Skills

HCI/UX Research	Mixed-Methods Studies, Experimental Design, Qualitative Coding(inductive, deductive, hybrid) Semi-Structured Interviews, Think-Aloud, Survey Design, Figma
XR Prototyping	Unity, OpenXR, C#, VR/AR Interaction, MRTK, Meta XR SDK, Android XR
Analytics	Gaze Analysis, Behavioral Log Analysis, Statistical Modeling, Data Visualization, Python, Pandas, ATLAS.ti
AI & ML	PyTorch, TensorFlow, Scikit-learn, Multi-Agent Systems, LLM Integration, RAG, LangGraph

Leadership and Service

President, Student Council *January 2026 — Present*
Center for Human-Computer Interaction at Virginia Tech (CHCI@VT), Blacksburg, VA
- Led council operations spanning faculty/student communication, event planning, and community building.
- **Hack & Yak:** Organized informal gatherings to foster collaboration between HCI students and faculty.
- **Cross-Departmental Workshops:** Collaborated with VT Libraries and SAIG to deliver workshops on AI-assisted research, R, and JMP statistical analysis.
- **Conference Practice Talks:** Facilitated presentation prep sessions for students submitting to venues such as CHI.
- **CHCI Research Symposium (April 2025):** Organized a symposium with **22 presentations** and **75 attendees**, providing a platform for peer and faculty feedback on student research.

Judge *March 2026*
AWS×Virginia Tech CS Careers Hackathon: Evaluated student projects; provided feedback across competing teams.

CS Senior Seminar Panelist *October 2025*
Shared insights on graduate school experiences, academic and professional development, and the application process with undergraduate students at Virginia Tech.

Reviewer
IEEE International Symposium on Virtual Reality and 3D User Interfaces(VR) 2025, 2026
IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2025, 2026
ACM Conference on Intelligent User Interfaces (ACM IUI) 2025
International Conference on Extended Reality (ICXR) 2026